

Initial Experiments: Extending the MVP Simulator

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Extends the MVP simulator (`docs/mvp/mvp_specification.tex`) by enriching the state space, action space, and reward function, moving toward the full MDP defined in `docs/design/initial_design.tex`.

1 State Space Extension

References `docs/design/initial_design.tex` §3 and Issue #124. *Implementation: PR 1b (feat/1b-states) — merged.*

1.1 Enriched StateView

The MVP dataclass currently holds four fields:

```
StateView(activity, day, step_of_day, steps_per_day)
```

PR 1b adds four optional float fields (all defaulting to `None` for MVP backward compatibility):

```
steps, weight, time_of_day, day_of_week
```

The transition ABC signature changes from `transition(state: str, action: str) -> str` to `transition(state: StateView, action: str) -> StateView`. The reward ABC signature changes from `reward(state: str, ...)` to `reward(state: StateView, ...)`.

When the config includes a `state_dynamics` block, `RuleBasedTransition` enters extended mode and evolves steps and weight according to the following equations:

$$\text{tod} = \text{step_of_day} \times \frac{24}{\text{steps_per_day}} \quad (1)$$

$$\text{dow} = \text{day} \bmod 7 \quad (2)$$

$$\text{steps_mean} = \text{response_multiplier}[a] \times \text{tod_modulation}[\text{tod}] \times \text{dow_modulation}[\text{dow}] \quad (3)$$

$$\text{steps}_{t+1} = \max(0, \text{steps}_t + \text{steps_mean} + \mathcal{N}(0, \sigma_{\text{steps}})) \quad (4)$$

$$\Delta \text{steps} = \text{steps}_{t+1} - \text{steps}_t \quad (5)$$

$$\text{weight}_{t+1} = \text{weight}_t + \text{meal_effect}[\text{tod}] + \text{weekend_boost} \cdot \mathbb{1}_{\text{dow} \in \{0,6\}} + \text{steps_coefficient} \cdot \Delta \text{steps} + \mathcal{N}(0, \sigma_{\text{weight}}) \quad (6)$$

The config schema defines two Pydantic models:

- **StepsDynamicsConfig**: `response_multiplier` (per-action step response), `tod_modulation` (hour-of-day gain, default 1.0 for missing hours), `dow_modulation` (day-of-week gain, default 1.0), and `noise_std` (non-negative, default 50).

- `WeightDynamicsConfig`: `meal_effect` (per-hour weight change), `weekend_boost` (additional weight gain on Sat/Sun, default 0.05), `steps_coefficient` (weight loss per step, default -10^{-4}), and `noise_std` (non-negative, default 0.05).

The experiment config is at `docs/initial_experiments/configs/1b_states_only.yaml`. It uses the 2 MVP actions with the above dynamics.

1.2 Environment Evolution (PR 1b)

`Environment.reset()` initialises `steps` and `weight` from the config's `initial_steps` (default 0.0) and `initial_weight` (default 70.0). `time_of_day` and `day_of_week` are computed deterministically from the step counter.

After each `step()` call, the environment overwrites the deterministic calendar fields (`day`, `step_of_day`, `time_of_day`, `day_of_week`) via `dataclasses.replace`. The transition model does not set these fields — they are the environment's responsibility.

1.3 Open Questions

- The `tod_modulation` keys are sparse (`{0, 6, 12, 18, 23}`). Missing hours default to 1.0, meaning most steps receive no time-of-day modulation. This may be refined with higher-resolution data.
- Steps and weight are unnormalised. Normalisation may be needed when these fields are used as agent context features.

2 Action Space Extension

References `docs/design/initial_design.tex` §3 and Issue #126.

2.1 ActionConfig

2.2 Transition Model Updates

2.3 Agent Compatibility

2.4 Open Questions

3 Reward Function Extension

References `docs/design/initial_design.tex` §3 and Issue #132.

3.1 Multi-Timescale Reward

3.2 Reward Composition

3.3 Synthetic Clinical Data

3.4 Open Questions

4 Combined Results

Results from the pairwise and full-integration experiments will appear here once all three extension PRs (1a, 1b, 1c) have been merged. Currently available:

- `docs/initial_experiments/configs/1b_states_only.yaml` (PR 1b, 2 actions, 4 continuous states, simple reward) — see §1.

PR 1b Results (50 seeds, all MVP agents).

Agent	Total Reward	Per Step	Weight (kg)	Steps
Standard TS	161.8 ± 15.9	0.3596	68.57 ± 1.22	57977 ± 5667
Contextual TS	178.8 ± 15.8	0.3974	69.50 ± 1.06	48702 ± 5210
Standard EG	155.3 ± 29.0	0.3452	68.80 ± 1.82	55728 ± 14096
Contextual EG	172.7 ± 26.9	0.3837	69.63 ± 1.62	47355 ± 10842
Standard UCB	163.6 ± 15.6	0.3635	68.34 ± 1.19	60253 ± 5517
Contextual UCB	179.2 ± 17.4	0.3982	69.51 ± 1.16	48626 ± 6131
Standard DEC	157.2 ± 25.6	0.3494	68.70 ± 1.70	56673 ± 13265
Contextual DEC	177.0 ± 28.8	0.3934	69.64 ± 1.45	47318 ± 9436
Random	138.3 ± 12.2	0.3073	70.31 ± 1.04	40630 ± 1832

Over 90-day episodes (450 steps) across 50 seeds, contextual agents consistently outperform their non-contextual counterparts by ~ 10 – 15 reward points. Contextual UCB achieves the highest mean reward (179.2 ± 17.4) while contextual TS and DEC follow closely. Weight drops from 70.0 kg across all agents (68.3–69.6 kg), driven by the cumulative meal effects and the negative `steps_coefficient`. Total steps range from 47k to 60k; contextual agents take fewer steps (47k–49k) yet achieve higher rewards, suggesting they learn to prefer `nudge` at higher-modulation hours. The per-step reward gap between contextual (~ 0.39) and non-contextual (~ 0.35) agents confirms that activity-based conditioning improves policy quality even with simple transition dynamics.